THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241
Introduction to Mathematical Analysis
Tutorial 2 Solution

Problem 1. Compute the limits:

(a) $\lim_{n\to\infty}\left(\frac{1}{2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{n\cdot(n+1)}\right)$

(b) $\lim_{n\to\infty}\sqrt[n]{\left(1-\frac{1}{n}\right)}$

(c) $\lim_{n\to\infty}\left(\frac{1}{n^2}+\frac{1}{(n+1)^2}+\cdots+\frac{1}{(2n)^2}\right)$

(d) $\lim_{n\to\infty}\left(\frac{1}{\sqrt{(n^2+1)}}+\frac{1}{\sqrt{(n^2+2)}}+\cdots+\frac{1}{\sqrt{(n^2+n)}}\right)$

Solution 1. (a) For $k\in\mathbb{R}$, $k\neq 0,-1$,

$$\frac{1}{k(k+1)}=\frac{1}{k}-\frac{1}{k+1}. \tag{1}$$

Therefore,

$$\begin{aligned}&\frac{1}{2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{n\cdot(n+1)}\\ =&1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots+\frac{1}{n}-\frac{1}{n+1}\\ =&1-\frac{1}{n+1}.\end{aligned}$$

By the basic operation of limit and $\lim\frac{1}{n}=0$ (proved in Example 1.7 in Lecture note), we have

$$\lim\frac{1}{2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{n\cdot(n+1)}=\lim 1-\frac{1}{n+1}=1. \tag{2}$$

(b) For $\sqrt[n]{a}$, $a<1$, we have $\sqrt[n]{a}<\sqrt[n]{1}=1$, $\sqrt[n]{a}>a^1=a$. Therefore,

$$1-\frac{1}{n}\leq\sqrt[n]{1-\frac{1}{n}}\leq 1.$$

By $\lim(1-\frac{1}{n})=1$ and squeez theorem, we have

$$\lim\sqrt[n]{1-\frac{1}{n}}=1. \tag{3}$$

(c) The n terms in the sum are bounded by $\frac{1}{(2n)^2}$ and $\frac{1}{n^2}$. Therefore,

$$\frac{1}{4n} = \frac{n}{(2n)^2} \le \frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2} \le \frac{n}{n^2} = \frac{1}{n}, \tag{4}$$

i.e.,

$$\lim_{n\to\infty} \Big(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2}\Big) = 0. \tag{5}$$

(d) Similar to (c), the terms in sequence can be bounded by:

$$\frac{n}{\sqrt{n^2+n}} \le \frac{1}{\sqrt{(n^2+1)}} + \frac{1}{\sqrt{(n^2+2)}} + \cdots + \frac{1}{\sqrt{(n^2+n)}} \le \frac{n}{\sqrt{n^2+1}}. \tag{6}$$

The limits of LHS and RHS are:

$$\lim \frac{n}{\sqrt{n^2+n}} = \lim \frac{1}{\sqrt{1+\frac{1}{n}}} = 0, \tag{7}$$

$$\lim \frac{n}{\sqrt{n^2+1}} = \lim \frac{1}{\sqrt{1+\frac{1}{n^2}}} = 0. \tag{8}$$

Therefore, by squeeze theorem,

$$\lim \frac{1}{\sqrt{(n^2+1)}} + \frac{1}{\sqrt{(n^2+2)}} + \cdots + \frac{1}{\sqrt{(n^2+n)}} = 0.$$

Problem 2. Let $a_1, a_2, \ldots, a_m$ be m positive numbers, prove that

$$\lim_{n\to\infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} = \max\{a_1, a_2, \cdots, a_m\}.$$

Solution 2. Denote $A := \max\{a_1, a_2, \cdots, a_m\}$. Then

$$A = \sqrt[n]{A^n} \le \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} \le \sqrt[n]{mA^n} = A\sqrt[n]{m}. \tag{9}$$

Notice that

$$1 \le \sqrt[n]{m} \le \sqrt[n]{n}, \tag{10}$$

and $\lim_{n\to\infty} \sqrt[n]{n} = 1$ (Theorem 5.2). By squeeze theorem, we have

$$\lim \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} = A.$$

Problem 3. (a) Let $\{a_n\}$ and $\{b_n\}$ be sequences. Prove that if $\lim_{n\to\infty} a_n = 0$ and if $\{b_n\}$ is bounded, then

$$\lim_{n\to\infty} a_n b_n = 0.$$

(b) Suppose now that $\lim_{n\to\infty} a_n = L \neq 0$ and that $\{b_n\}$ is bounded. What can we conclude about the convergence of the sequence $\{a_n b_n\}$?

Solution 3. (a) Since $\{b_n\}$ is bounded, there exists $M > 0$ such that

$$|b_n| \leq M \qquad \text{for all } n \in \mathbb{N}.$$

Now, for any $\epsilon > 0$, consider the number $\epsilon/M > 0$. Since $\{a_n\}$ converges to 0, there exists $N \in \mathbb{N}$ such that for any $n \in \mathbb{N}$, if $n \geq N$, then

$$|a_n| < \frac{\epsilon}{M}.$$

It follows that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $n \in \mathbb{N}$, if $n \geq N$, then

$$|a_n b_n| = |a_n||b_n| \leq M|a_n| < M \cdot \frac{\epsilon}{M} = \epsilon.$$

(b) There is nothing we can conclude from this situation, as there are examples, say

$$\{a_n\} = \{b_n\} = \{1, 1, 1, 1, \ldots\},$$

in which the hypotheses are satisfied and the sequence $\{a_n b_n\}$ converges, while there are also examples, say

$$\{a_n\} = \{1, 1, 1, 1, \ldots\} \qquad \text{and} \qquad \{b_n\} = \{(-1)^n\},$$

in which the hypotheses are satisfied but the sequence $\{a_n b_n\}$ diverges.

Problem 4. Suppose $x_1 = \sqrt{2}$, $x_{n+1} = \frac{1}{x_n+2}$, $(n = 1, 2, 3\ldots)$, prove that the sequence $\{x_n\}$ is convergent, and find the limit.
Hint: If $0 < x_n < \sqrt{2} - 1$, then $x_{n+1} > \sqrt{2} - 1$; if $x_n > \sqrt{2} - 1$, then $0 < x_{n+1} < \sqrt{2} - 1$.

Solution 4. If $0 < x_n < \sqrt{2} - 1$, then

$$x_{n+1} = \frac{1}{x_n + 2} > \frac{1}{\sqrt{2} + 1} = \frac{\sqrt{2} - 1}{(\sqrt{2})^2 - 1} = \sqrt{2} - 1.$$

If $x_n > \sqrt{2} - 1$, then

$$x_{n+1} = \frac{1}{x_n + 2} < \frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1. \tag{11}$$

With $x_1 = \sqrt{2} > \sqrt{2} - 1$, we have $x_{2k-1} > \sqrt{2} - 1$ and $0 < x_{2k} < \sqrt{2} - 1$, $k \in \mathbb{N}$. We prove the convergence of the two subsequences $\{x_{2k}\}$ and $\{x_{2k-1}\}$ seperately.

$$x_{n+1} = \frac{1}{x_n + 2} = \frac{1}{\frac{1}{x_{n-1}+2} + 2} = \frac{x_{n-1} + 2}{2x_{n-1} + 5}. \tag{12}$$

Therefore,

$$\begin{aligned} x_{2k+1} - x_{2k-1} =& \frac{x_{2k-1}+2}{2x_{2k-1}+5} - x_{2k-1} \\ =& \frac{-2x_{2k-1}^2 - 4x_{2k-1}+2}{2x_{2k-1}+5} = \frac{-2(x_{2k-1}-(\sqrt{2}-1))(x_{2k-1}-(-\sqrt{2}-1))}{2x_{2k-1}+5} < 0. \end{aligned}$$

Similarly

$$x_{2k} - x_{2k-2} = \frac{-2(x_{2k-2}-(\sqrt{2}-1))(x_{2k-2}-(-\sqrt{2}-1))}{2x_{2k-2}+5} > 0.$$

Hence, $\{x_{2k}\}$ is a bounded increasing sequence and $\{x_{2k-1}\}$ is a bounded decreasing sequence. By MCT, they are both convergent. For $\{x_{2k}\}$, denote the limit as L, then L satisfies

$$\begin{aligned} & L = \frac{L+2}{2L+5} \\ \iff \quad & L = \pm\sqrt{2} - 1. \end{aligned}$$

Since $x_{2k} > 0$, the limit is positive, $\lim x_{2k} = \sqrt{2} - 1$. Similarly, one can prove $\lim x_{2k-1} = \sqrt{2} - 1$. Therefore, $\lim x_n = \sqrt{2} - 1$.

Problem 5. (a) Suppose $a_n > 0$, $(n = 1, 2, \cdots)$, and $\lim \frac{a_n}{a_{n+1}} = l > 1$, prove that $\lim a_n = 0$.

(b) Prove that the limit $\lim_{n\to\infty} \frac{n^2}{3^n}$ exists and find the limit.

Solution 5. (a) Since $l > 1$, there exists $r \in (1, l)$. Set $\varepsilon = l - r > 0$, then there exists $N > 0$, such that $\forall n \geq N$, we have

$$\begin{aligned} & \left|\frac{a_n}{a_{n+1}} - l\right| \leq \varepsilon, \\ \iff \quad & l - \epsilon \leq \frac{a_n}{a_{n+1}} \leq l + \epsilon. \end{aligned}$$

That is,

$$a_{n+1} \leq a_n(l-\varepsilon)^{-1} = r^{-1}a_n, \quad n \geq N \tag{13}$$

By applying 13 repeatedly, we have

$$0 < a_{n+1} \leq r^{-n-1+N} a_N.$$

Apply squeeze theorem and $\lim r^{-n} = 0$, $(r > 1)$, we have $\lim a_n = 0$.

(b) Define $a_n = \frac{n^2}{3^n}$, then

$$\frac{a_n}{a_{n+1}} = \frac{n^2}{3^n} \cdot \frac{3^{n+1}}{(n+1)^2} = \frac{n^2}{(n+1)^2} \cdot 3. \tag{14}$$

Then $\lim \frac{a_n}{a_{n+1}} = \lim \frac{n^2}{(n+1)^2} \cdot 3 = 3 > 1$. From (a) we know that $\lim a_n = 0$.

Food for Thought. Find the limits $\lim_{n\to\infty}\frac{3^n}{n!}$, $\lim_{n\to\infty}\frac{n!}{n^n}$ and compare the 'ultimate behaviour' of n^a, b^n, $n!$ and n^n, $(a, b > 1)$.

Problem 6. Find the limits:

(a)

$$\lim_{n\to\infty}\frac{\sum_{p=1}^{n} p!}{n!}$$

(b)

$$\lim_{n\to\infty}\big((n+1)^\alpha - n^\alpha\big), \quad 0 < \alpha < 1$$

Solution 6. (a)

$$\begin{aligned}1! + 2! + \cdots n! \leq &(n-2)\cdot(n-2)! + (n-1)! + n!\\ \leq &(2n-3)\cdot(n-2)! + n!\end{aligned}$$

That is

$$0 < \frac{\sum_{p=1}^{n} p!}{n!} < \frac{(2n-3)\cdot(n-2)!}{n!} + 1 = \frac{2n-3}{n(n-1)} + 1.$$

Appy the squeeze theorem, we have $\lim_{n\to\infty}\frac{\sum_{p=1}^{n} p!}{n!} = 1$.

(b)

$$\begin{aligned}(n+1)^\alpha - n^\alpha =& n^{\alpha-1} + \big((n+1)^\alpha - n^\alpha - n^{\alpha-1}\big)\\ =& n^{\alpha-1} + n^{\alpha-1}\Big(\frac{(n+1)^\alpha}{n^\alpha}\cdot n - n - 1\Big)\\ \leq& n^{\alpha-1} + n^{\alpha-1}\Big(\frac{n+1}{n}\cdot n - n - 1\Big)\\ =& n^{\alpha-1}.\end{aligned}$$

Therefore,

$$0 < (n+1)^\alpha - n^\alpha < n^{\alpha-1}. \tag{15}$$

By squeeze theorem and $\lim n^{\alpha-1} = 0$, $(0 < \alpha < 1)$, we have $\lim(n+1)^\alpha - n^\alpha = 0$.